**PY211IB**

**USN** | | | | | | | | | | |

# RV COLLEGE OF ENGINEERING®

**(An Autonomous Institution affiliated to VTU)**

**I / II Semester B. E. Regular / Supplementary Examinations Aug-2024**

**COMMON TO ME, AS, IM & CH**

## CLASSICAL PHYSICS FOR ENGINEERS

***Time: 03 Hours*** ***Maximum Marks: 100***

***Instructions to candidates:***

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 & 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10, and 11 lab components (compulsory).
3. Handbook of Physics is allowed.

**PART A**

| | | | M | BT | CO |
|---|---|---|---|---|---|
| 1 | 1.1 | What is sharpness of resonance? | 01 | 1 | 1 |
| | 1.2 | List any two characteristics of $SHM$. | 01 | 1 | 1 |
| | 1.3 | Write the limitations of Poisson's ratio. | 01 | 1 | 1 |
| | 1.4 | Define neutral layer of a cantilever. | 01 | 2 | 1 |
| | 1.5 | State zeroth law of thermodynamics. | 01 | 1 | 1 |
| | 1.6 | How friction generates irreversibility in a thermodynamic process? | 01 | 2 | 2 |
| | 1.7 | Give an example for non-Newtonian fluid. | 01 | 1 | 1 |
| | 1.8 | What is cavitation? | 01 | 1 | 1 |
| | 1.9 | Name any two surface characterization techniques. | 01 | 1 | 1 |
| | 1.10 | Mention any two disadvantages of AFM. | 01 | 1 | 1 |

**PART B**

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | What are damped vibrations? Establish the differential equation of motion for a damped harmonic oscillator and obtain an expression for displacement. Discuss the case of low damping. | 10 | 2 | 2 |
| | b | A vibrating system of natural frequency 500 cylcies/s is forced to vibrate with a periodic force/unit mass of amplitude $100 \times 10^{-5}\ N/kg$ in the presence of a damping /unit mass of $0.01 \times 10^{-3}$ rad/s. Calculate the maximum amplitude of vibration of the system. | 04 | 2 | 3 |
| | | | | | |
| 3 | a | Derive an expression for the moment of the couple required to twist one end of a cylinder, when the other end is fixed. Hence arrive at the periodic time of a torsional pendulum. | 10 | 3 | 1 |
| | b | A wire of length $1.5m$ and radius $2mm$ is fixed at one end and twisted at the other end by 10°. Find the angle of shear on the surface of the wire? | 04 | 2 | 3 |
| | | **OR** | | | |
| 4 | a | Show that shearing stress is equivalent to an equal linear tensile stress and an equal linear compressive stress at right angles to each other. | 10 | 2 | 1 |
| | b | A wire of $0.3mm$ diameter elongates by 1mm when stretched by a force of $0.33\,kg$ weight and twists through 90° when equal and opposite torques of $2.15 \times 10^{-5}$ Nm are applied at its ends. Determine the poison's ratio of the wire. | 04 | 2 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| 5 | a | Explain first law of Thermodynamics. Discuss the construction and working of Six's maximum and minimum thermometer and Discuss scales of temperature. | 10 | 1 | 2 |
| | b | When a value of a sealed bottle is opened, atmospheric air rushes in with a process of $101.325kPa$ and $0.6m^3$ of air is measured at atmospheric pressure. Calculate the work done by air. | 04 | 2 | 3 |
| | | **OR** | | | |
| 6 | a | Define adiabatic expansion of gas. Derive an expression for the adiabatic work done by an external force on the ideal gas by deriving the adiabatic equation of state. | 10 | 1 | 1 |
| | b | The pressure of gas in a pipe line is measured with a mercury manometer having one limb open to the atmosphere. If the difference in the height of mercury in the two limbs is $562mm$, calculate the gas pressure. The barometer reads $761mm\,Hg$, the acceleration due to gravity is $\rho o = 9.79m/s^2$ and the density of mercury is 13,640 $kg/m^3$ | 04 | 3 | 3 |
| 7 | a | What are ultrasonic waves? Explain principle, construction and working of ultrasonic interferometer with diagram. | 10 | 2 | 2 |
| | b | A 0.6mm diameter glass tube is inserted into water at $20°C$ in a cup. Determine the capillary rise of water in the tube. Given the surface tension of water at $20°C$ is $0.073N/m$. | 04 | 1 | 3 |
| | | **OR** | | | |
| 8 | a | Arrive at continuity equation in integral form for the flow of an ideal fluid through a tube. Explain the difference between stream line and Streak line. | 10 | 2 | 1 |
| | b | An ultrasonic interferometer is operated with alternating current of $2 \times 10^6$ Hz. With a liquid of density $996.45\,kg/m^3$ placed in its measuring cell, two consecutive micrometer readings for minimum current are found to be $1.100mm$ and $1.349\,mm$. Determine adiabatic compressibility of the liquid in the cell. | 04 | 1 | 3 |
| | | | | | |
| 9 | a | With a neat labeled diagram, explain principle, construction and working of Transmission Electron Microscope ($TEM$). | 09 | 2 | 2 |
| | b | Derive Bragg's law. | 05 | 2 | 1 |
| | | **OR** | | | |
| 10 | a | Explain the construction and working of Bragg's diffractometer with neat diagram. | 08 | 1 | 2 |
| | b | Explain two types of hardness testing techniques for the material. | 06 | 2 | 2 |
| | | | | | |
| 11 | a | Write the condition for diffraction of light to be observed. Write the experimental procedure to determine wavelength of given laser beam using a diffraction grating from the following data.<br>Distance between the grating and screen $d = 90cm$<br>Grating constant=$5.05 \times 10^{-5}$m | 10 | 3 | 4 |

| $Diffraction\ order$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $Distance\ X_n\ (cm)$ | 1.3 | 2.6 | 3.9 | 4.9 | 6.2 | 7.9 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| b | Following is the data in an experimental determination of Fermi energy of copper. Plot a graph of resistance as function of temperature and calculate the Fermi energy of copper.<br>Given length of wire= $15m$<br>Density of wire= $8960Kg/m^3$ Radius of wire= $0.1mm$ | | | | | | | | | | 10 | 2 | 4 |
| | *Temp in centigrade* | 24.6 | 80 | 75 | 65 | 60 | 55 | 50 | 45 | 40 | | | |
| | *Resistance in ohms* | 10 | 12.1 | 11.9 | 11.8 | 11.6 | 11.4 | 11.2 | 11 | 10.6 | | | |